

# Chandohasam: Telugu Metrical Poetry Generation with an Exact Metrical Controller and Text Diffusion

Samvaran Kashyap Rallabandi, Mahesh Emani, Radhe Shyam Salopanthula  
IIIT Hyderabad

{samvaran.rallabandi, mahesh.emani, radheshyam.s}@research.iiit.ac.in

## Abstract

A Telugu metrical poem (*padam*) must satisfy hard constraints on syllable weight (*gaṇa*), caesura (*yati*) and line-initial rhyme (*prāsa*) while conveying a meaning. We treat *padam* generation as constrained generation: validity can be decided exactly; meaning and quality cannot. We report four results. (1) A rules-as-data controller for 37 metres: a minimal labelled automaton (DAWG) of 1,431 states that accepts exactly the 370,315 legal lines, with *yati* and *prāsa* engines; it identifies 7,355 of 7,365 labelled poems of Pothana’s *Bhāgavatamu* and none as a wrong metre. (2) Diagnostics: free generation from three open models gives 0 of 1,665 poems in metre, and a 262K subword vocabulary covers only 10.6% of the syllable types of Telugu verse. (3) Three constrained decoding strategies that put the controller inside autoregressive, block-diffusion and masked-diffusion generators: all 5,550 constrained poems are in metre, but the share of real words falls to 10.7–20.8% for subword models, while our 120M syllable-level diffusion model keeps 36.9%. (4) An audited corpus and a deterministic poetics suite derived from the *Kāvyālaṅkārasaṅgrahamu*. We close with a plan for fine-tuning and a 500-poem human study.

## 1 Introduction

Telugu *padams* are composed in metres that fix the weight of every syllable (*akṣara*) of a line, a caesura (*yati*) at which an *akṣara* must agree in sound with the line’s first *akṣara*, and a rhyme (*prāsa*) on the second *akṣara* of every line. Large language models do not satisfy these constraints. Prompted once per poem, Gemini 3.7 Flash wrote 255 poems, of which 91 (35.7%) are strictly valid under our engines (Appendix B) [E1]. Three open models (Gemma-4 E4B, Gemma-4 26B-A4B and DiffusionGemma 26B-A4B) write 0

of 1,665 poems in metre when left free (§6.2) [E22]. The difficulty is not that the rules are unknown. They are codified in the prosody treatises, such as the seventeenth-century *Appakavīyam* (Appakavi, 1966), and are exact.

### Problem statement.

- **Poems and scansion.** Let  $\Sigma$  be the set of *akṣaras*. A poem is a sequence of *pādas*  $y = (y_1, \dots, y_P)$ ,  $y_p \in \Sigma^*$ . Scansion maps an *akṣara* sequence to a *set* of weight strings over  $\{u, l\}$  (*guru*, *laghu*). It is a set because a light *akṣara* before a conjunct is heavy, and some *akṣaras* have two accepted readings.
- **Metres.** A metre  $M$  fixes the number of *pādas* and a slot  $s(p)$  for each; a finite language  $L_s \subseteq \{u, l\}^*$  for each slot, given by its *gaṇa* grammar; *yati* seats  $K_s$ , at which the *akṣara* must be in sound-class harmony (*maitri*) with the *pāda*’s first *akṣara* under a rule set  $R_Y$ ; and a *prāsa* relation  $R_P$  on the second *akṣara* of every *pāda*.
- **Validity.**  $y$  is *valid* for  $M$ ,  $\text{Valid}_M(y)$ , when every *pāda* has a reading in  $L_{s(p)}$  and the *yati* and *prāsa* relations hold.
- **The task.** Given a meaning  $m$  (a prose *bhāvam* or a topic) and a metre  $M$ , generate  $y$  with  $\text{Valid}_M(y)$  that expresses  $m$  in fluent, real Telugu and has the qualities the tradition values (the *guṇas*).

Validity is decidable exactly, because every  $L_s$  is finite and therefore regular. Meaning and quality are not.

**Approach** (Figure 1). We separate the decidable part from the rest. A deterministic controller decides validity exactly and is used *inside* the decoding loop, so every generated poem is valid by construction; a generator supplies the language and the meaning; and a separate evaluation suite measures what validity does not. Our generator of choice is a masked diffusion language model with one token per *akṣara*: its units are the metre’s units,

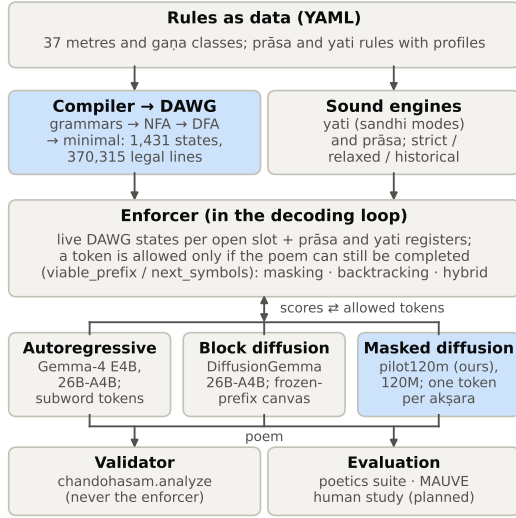


Figure 1: System overview. Metre rules (YAML) compile into the DAWG and the yati and prāsa engines. An enforcer consults them inside the decoding loop of any generator. A separate validator and the evaluation suite score the finished poem.

and diffusion lets an akṣara be chosen with both of its neighbours in view.

**Research questions and contributions.** **RQ1:** Can validity be decided exactly for the metres of Telugu? We build and validate a rules-as-data controller for 37 metres (§4). **RQ2:** Why do pretrained models fail at it? The failure lies in their units, not in their exposure to the rules (§6.1). **RQ3:** Can validity be guaranteed across generator architectures, and at what cost? We introduce three constrained decoding strategies (masking only, masking with backtracking, hybrid), use them in three kinds of generator, and measure the cost (§5, §6.2). **RQ4:** How can we measure what validity does not capture? We audit a 15,415-record corpus with a control for every audit metric (§3) and build a deterministic poetics suite from the *Kāvyaṭaṅkārasaṅgrahamu* (Rāmarājabhūṣaṇuḍu, 1920) (§6.3). Whether fine-tuning closes the meaning gap under the constraint is the remaining work (§8).

## 2 Related Work

**Poetry under form constraints.** Form has been controlled by two routes. The *soft route* learns or prompts for form and filters failures: Deep-speare models metre jointly with language for English sonnets (Lau et al., 2018); SongNet adds format, position and segment symbols (Li et al., 2020); PoELM trains on non-poetic text

with codes for length and end rhyme and keeps samples that pass syllable-count and rhyme filters, as only 31% (Spanish) and 23% (Basque) of raw samples meet the constraints (Ormazabal et al., 2022); and a Bangla adaptation adds control tokens for rhyme, mātrā count, word count and line boundaries, then filters and re-ranks (Amina et al., 2025). These controls are aggregate, and validity comes from filtering, not from the model. The *hard route* constrains decoding: finite-state acceptors for rhythm and rhyme (Ghazvininejad et al., 2016), lexically constrained beam search (Hokamp and Liu, 2017; Lu et al., 2021), grammar-constrained decoding (Geng et al., 2023) and finite-state indexing of the vocabulary (Willard and Louf, 2023). For Sanskrit, Chandomitra translates English prose into *anuṣṭubh* verse with 99.86% metrically valid output under constrained decoding, but human raters score the semantic coherence of the constrained system at 2.13 out of 5, against 3.78 for an unconstrained instruction-tuned model (Jagadeeshan et al., 2026a); Pingala reaches near-perfect metrical conformity by biasing decoding towards complete words at pāda boundaries (Jagadeeshan et al., 2026b). Meaning under the constraint is the open problem. PoetryDiffusion pairs continuous diffusion over word embeddings with a separately trained BERT-based controller for format, rhyme and tone (Hu et al., 2024). Our controller is exact instead, and our decoding strategies serve autoregressive and diffusion generators alike.

**Computational prosody.** Finite-state scansion exists for English (Agirrezabal et al., 2013) and for Ancient Greek hexameter (Schumann et al., 2022). Sanskrit metre identification compares a line’s weight string with known patterns (Melnad et al., 2015; Rajagopalan, 2018; Terdalkar and Bhattacharya, 2023); Chandojñānam also uses the edit distance to the nearest pattern to suggest corrections (Terdalkar and Bhattacharya, 2023). For Telugu, Boddu and Boddu (2025) score existing poems against metre patterns and release the Chandassu dataset, which we audit and use. We differ in compiling every metre into one minimal automaton (Rabin and Scott, 1959; Moore, 1956; Daciuk et al., 2000; Blumer et al., 1985), which both decides validity and answers the prefix queries a decoder needs.

**Discrete diffusion language models.** Continuous diffusion on embeddings was first

used for controllable text (Li et al., 2022). Discrete diffusion over tokens was introduced by D3PM (Austin et al., 2021), and score-entropy discrete diffusion outperformed continuous diffusion language models at GPT-2 scale (Lou et al., 2024). Masked diffusion with a simple weighted cross-entropy is now competitive (Sahoo et al., 2024; Shi et al., 2024), needs no time input (Ou et al., 2025; Zheng et al., 2025), and scales (Nie et al., 2025). Block diffusion interpolates between autoregressive and diffusion decoding (Arriola et al., 2025). Remasking samplers let a model revise tokens it has already unmasked (Wang et al., 2025).

**Data, units and evaluation.** Pretraining text for Telugu comes from IndicCorp v2 and Sangraha (Doddapaneni et al., 2023; Khan et al., 2024). Corpus audits show that collected text must be checked before it is trusted (Kreutzer et al., 2022; Dodge et al., 2021), and deduplicating training data reduces memorisation and train–test overlap (Lee et al., 2022). Subword vocabularies split Telugu akṣaras (§6.1); our model uses one token per akṣara. MAUVE compares distributions of generated and human text (Pillutla et al., 2021); LLM judges rate text quality (Zheng et al., 2023; Liu et al., 2023); human protocols annotate poetry for aesthetic response (Haider et al., 2020). For Sanskrit, computational work has detected figures of sound (anuprāsa, yamaka) (Barbadikar and Kulkarni, 2024, n.d.) and of meaning (upamā) (Jadhav et al., 2025), and has analysed a poem through the schools of Sanskrit poetics (Sandhan et al., 2025). For English poetry, style has been measured from word frequency and sound (Kao and Jurafsky, 2012).

### 3 Data and its Audit

**Evaluation corpora.** Four files share one record format (Table 1): Pothana’s *Āndhra Mahābhāgavatamu*, from an online compilation that gives word glosses and meanings (Pothana, n.d.); Vemana’s *śatakam*; Kuchimanchi Timmakavi’s pure-Telugu *Rāmāyaṇamu*; and the Chandassu dataset (Boddu and Boddu, 2025), from its Kaggle release (Boddu, 2025). Together they hold 15,415 records, of which 12,704 are verse poems [E2]. Outside the Bhāgavatamu, meanings and word glosses are machine-written (Gemini batch runs), and every English meaning is a machine translation of the Telugu one.

**Audit.** We audited the files at five levels: inventory (level 0), prosodic integrity (1), meaning-to-verse length ratio (2), embedding similarity (3), and diversity, subword fit and duplicates (4). An audit metric can measure less than its name says (the similarity gate below is one), so every metric here carries a **control** that tests whether the number means what it claims.

- **The prosodic rules are hard to satisfy by chance** [E4]. Material from unrelated poems of the same metre passes 14–18% of yati seats and 0.0–3.7% of prāsa stanzas, against 73–93% and 93–100% for real verse.
- **The audit found real faults** [E3], such as backslashes inside words, lost pāda breaks and 142 Vemana poems mislabelled by a heuristic. Fixing them raised the share of fully valid poems from 72.4% to 83.5% (Vemana) and from 63.8% to 74.0% (Chandassu).
- **The original similarity gate measures translation consistency, not fidelity** [E5]. Telugu–English meaning pairs pass LaBSE  $\geq 0.65$  (Feng et al., 2022) at 90–98%, because one translates the other. Poem and Telugu meaning, the pair that bears on fidelity, separate right from wrong pairings (AUC 0.85–0.94) yet clear 0.65 for only 0.7–5.3% of poems, and an English-only encoder that cannot read Telugu “passes” 94–99% of those pairs. Fixed thresholds are not quality gates here.
- **Duplicates** [E6]. No poem appears in two files, but refrain lines (*makūṭa*) recur: 77.1% of Vemana poems share a whole pāda with another poem, so splits keep refrain groups together.

**Training resources.** Fine-tuning will use Telugu poems crawled from the internet and from blogs, with their meanings (machine-written where none is given). The pretraining corpus has 4.98B tokens of Telugu from Sangraha, IndicCorp v2 and Wikipedia, without any document that shares a 20-token window with a known poem. The Bhāgavatamu is held out of fine-tuning (906 validation and 5,411 test poems).

### 4 The Metrical Controller

**Rules as data.** The rules live in YAML files, not in code: 37 metres and their gaṇa classes; 64 prāsa rules and 92 yati rules, each with a status that profiles (strict, relaxed, historical) accept or reject; and sandhi modes for yati (off, a one-sided hypothesis, vowel-only).

| file        | records<br>(verse) | metre<br>label     | valid, relaxed<br>(strict) | yati seat:<br>real / chance | prāsa stanza:<br>real / chance | AUC  |
|-------------|--------------------|--------------------|----------------------------|-----------------------------|--------------------------------|------|
| Bhāgavatamu | 10,066 (7,366)     | compilation        | 91.2% (88.7%)              | 88.5% / 17.3%               | 99.8% / 0.2%                   | 0.93 |
| Vemana      | 1,164 (1,164)      | heuristic / engine | 83.5% (82.6%)              | 92.5% / 16.6%               | 95.5% / 0.0%                   | 0.94 |
| Kuchimanchi | 1,653 (1,642)      | engine             | 90.6% (88.7%)              | 73.5% / 14.1%               | 99.9% / 3.7%                   | 0.85 |
| Chandassu   | 2,532 (2,532)      | dataset            | 74.0% (69.9%)              | 87.4% / 16.4%               | 98.8% / 0.1%                   | 0.93 |

Table 1: The evaluation corpora and their audit. *Valid*: gaṇa, prāsa and yati all hold under the relaxed (strict) profile. *Chance*: pass rate of material from unrelated poems of the same metre (relaxed). *AUC*: LaBSE poem–meaning pairs against shuffled pairs [E2–E5].

**From rules to automaton** (Algorithm 1, Appendix). (1) Each metre is written as a prosodic grammar: poem  $\rightarrow$  lines  $\rightarrow$  gaṇa classes  $\rightarrow$  gaṇas  $\rightarrow$  u/I. (2) The grammar is flattened into one right-linear grammar per (metre, line slot), 40 in all; in strict form, each is an NFA (Hopcroft and Ullman, 1979). (3) The union of the slot NFAs (1,171 states) is determined by subset construction (Rabin and Scott, 1959) into a DFA of 1,573 states, whose accepting states carry their (metre, slot) labels. (4) Moore partition refinement (Moore, 1956), started from the partition by label set, gives the minimal labelled automaton: 1,431 states and 2,514 edges [E7]. The result is a minimal, deterministic, acyclic automaton (Daciuk et al., 2000). We call it a DAWG (directed acyclic word graph), the name Blumer et al. (1985) gave to an automaton that recognises the subwords of a text. It accepts exactly the 370,315 legal lines and builds in under half a second. Minimisation matters: a trie of these lines would need 1,417,226 states. Figure 2 shows the idea on a toy slot.

**Queries.** identify walks a poem’s scansion lattice, so open readings (*vikalpa*) are tried together rather than enumerated, and returns every metre and slot that accept and where each rejected metre dies. *viable\_prefix* and *next\_symbols* return the symbols after which some metre slot can still be completed (co-reachable label sets): the query a decoder asks at every step. *segment* returns the gaṇa division and the yati positions that the sound engines check. 16,979 legal lines (4.6%) fit more than one metre, 16,416 of them shared by *sīsamu* and *taruvōja*; they are reported as ambiguous and never resolved by guessing.

**Validation** (Appendix Table 4). On canonical verse, the DAWG identifies 7,355 of the 7,365 Bhāgavatamu poems that the compilation labels with a catalogue metre (99.86%) as that metre, none as a different metre, and 10 as none [E8].

Against the per-akṣara weights distributed with the Chandassu dataset (Boddu and Boddu, 2025), 99.10% of akṣaras agree [E9]. 541 tests pass, including an equivalence check between every grammar and its automaton by full enumeration, and agreement with a brute-force identification oracle on valid, mutated and random stanzas [E10]. **Yati** is where the printed text and the rule diverge most. Pothana’s poems pass yati at 63.6% when no sandhi is assumed, and at 90.7% when a one-sided sandhi hypothesis is allowed [E11]: sandhi, which the printed text does not show, explains most apparent yati faults.

## 5 Integrating the Controller with Generators

**The enforcer** (Figure 1; Algorithm 2, Appendix). At each step the enforcer keeps the live states of the metre’s automaton, built as the DAWG is, and two registers for prāsa and yati. A candidate token is allowed if, after its akṣaras, some slot can still be completed and the registers still admit a completion; all other tokens are masked. The weight of an akṣara stays pending until the next onset is known, since a following conjunct makes it heavy. Because liveness is exact, no run can reach a dead end and every finished poem is valid; evaluation uses the separate analyser, never the decoder. We introduce three decoding strategies on the enforcer (Algorithm 3): *masking only*, which samples from the allowed tokens; *masking with backtracking*, which rewinds to a checkpoint and raises the temperature when the allowed set or its mass collapses; and *hybrid*, which re-ranks the allowed tokens towards completing the line.

**Autoregressive models.** For Gemma-4 E4B and 26B-A4B (Gemma Team, 2026; Google DeepMind, 2026b,a), the enforcer masks the next-token distribution. A subword token may span several akṣaras, and all of them are checked.



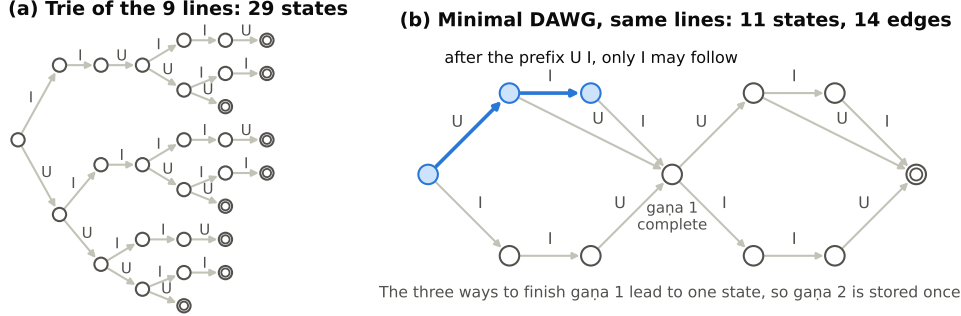


Figure 2: The DAWG idea on a toy slot of two gaṇas, each UU, IU or UI: (a) the trie of its 9 legal lines has 29 states; (b) merging states with equal futures leaves 11 states and 14 edges. After the prefix UI, only I may follow: this is the query the decoder asks.

The prompt states the metre’s rules, generated from the catalogue, and a topic.

**Block diffusion.** For DiffusionGemma 26B-A4B (NVIDIA, 2026), which samples block by block, the constraint acts through a frozen-prefix canvas: tokens committed under the mask stay fixed while the model redrafts the rest of the block. Two adaptations were needed: an empty thinking-channel prefill, and forbidding end-of-text inside the canvas.

**Our masked-diffusion model.** pilot120m is a 120M-parameter bidirectional transformer (12 layers, hidden size 768, rotary positions, SwiGLU), trained from scratch as a masked diffusion language model (Sahoo et al., 2024) with no time input (Ou et al., 2025); masking, unlike uniform corruption, marks which positions are still undecided, which the constraint needs. It saw 4.98B tokens of Telugu prose on a single 8 GB laptop GPU in 2 days 19 hours (test NELBO 1.783 nats per token) [E12]. Its tokenizer has 45,591 tokens and gives 99.8% of the akṣara types of our verse their own token [E13], so the constraint acts on single syllables and the template position equals the token position. Generation fills a 512-token canvas: the meaning is given unmasked, and the metre’s lines are decoded left to right within the line. Prāsa is checked by the compiled prāsa rules, and yati by an oracle that returns the yati engine’s exact verdict in its left context. An earlier yati table, used out of context, accepted pairs such as ు-ృ that no poem supports and reached 425 of 555 poems in metre; with the oracle and fixes to the sīsa half-break, couplet metres and prāsa, the decoder reaches 555 of 555 (§6.2) [E23].

## 6 Experiments

All results are on the common grid used for every model unless stated: 37 metres  $\times$  3 topics  $\times$  5 seeds, 555 poems per model and decoding mode. A poem is *in metre* when it is finished and its canonical gaṇa, strict prāsa and strict yati (no sandhi assumed) all hold.

### 6.1 Why pretrained models fail

The diagnostics use gemma-4-E2B-it and 200 Bhāgavatamu poems (25 from each of 8 metres) with their meanings.<sup>1</sup>

**Free generation fails at chance level.** Given the meaning and the metre’s rules, the model writes 0 of 200 poems in any metre, although 172 have four lines; 50.8% of akṣaras within the template’s length have the wrong weight, about what chance gives [E15]. In a logistic regression over steps, neither entropy nor distance from a word junction predicts a violation; only a rare token does.

**Weight is looked up, not computed.** A linear probe at each of the 36 layers predicts the current akṣara’s weight. On the 3,938 tokens whose weight depends on the next akṣara, the best layer is the embedding layer (84.7%), below a lookup of the current akṣara (88.7%) and far below one that also sees the next onset (99.1%) [E16].

**Reading verse shows no sense of the metre.** Verse is harder than its own meaning (6.40 against 5.35 nats; 186 of 200 pairs); pāda starts are *less* surprising than other word starts; and prāsa and

<sup>1</sup>The Gemma tokenizer does not add <bos>. Without it, the mean surprisal of verse is 13.45 nats per token, above the uniform bound  $\ln |V| = 12.48$ ; with it, 6.40 [E14]. All analyses prepend <bos>.

yati seats do not become easier once their anchor is fixed [E14].

**Exposure bias is not the main cause.** Under teacher forcing the top choice has the template weight in 68.4% of slots (51.0% when free), so even with the poet’s own prefix it misses about a third [E15]. Bidirectional context does not help by itself: DiffusionGemma, which revises within a block, writes 0 of 555 poems in metre when free [E22].

**The units do not match.** The 262,144-token Gemma vocabulary has 475 single-akṣara tokens, covering 10.6% of the 3,612 akṣara types of our verse (81.9% of occurrences; ours: 99.8%) [E13], and 15.5–21.8% of its token boundaries in verse fall inside an akṣara [E17]. A model that emits two or three syllables at once cannot choose one syllable’s weight.

**Steering is not an alternative.** Register (verse against prose) is linearly separable from layer 1 (0.978) [E18], and per-metre directions mirror the symbolic distance between metres at 34 of 36 layers (peak Spearman 0.77) [E19]. But no metre-against-metre direction is consistent across poems (all 28 pairs below 0.16 coherence, under a 0.32 threshold) [E20], even on 357 edited contrast poems [E21]. Metre is encoded as an average, not as a direction to steer with, which motivates a hard constraint.

## 6.2 Form by construction, and its cost

Table 2 gives the main results; Appendix Table 5 gives every decoding strategy [E22–E25].

- **The constraint guarantees form.** All 4,995 constrained poems of the three Gemma models are in metre, and so are the 555 masked poems of pilot120m; free generation gives 0 of 555 for every model. Under the chandas distance (§6.3), free poems are *farther* from the requested metre than prose cut to the metre’s line lengths (mean skill  $-0.20$  to  $-0.33$ ) [E25]: the lines are too short.
- **The constraint has a lexical cost for subword models.** Words attested in held-out verse (topic words excluded) fall from 33.7–37.2% to 10.7–20.8% [E24]. The metre overrides the model most often at yati positions (52% of steps for E4B, 66% for DiffusionGemma) [E22]. A quarter to four-fifths of the poems reach a level of alliteration that 2.7% of real poems reach,

the 26B models build heavy syllables from rare clusters with ँ and ः, and 83–99% of the poems are rated obscure (real verse: 10%) [E25].

- **Syllable-level units preserve vocabulary.** Under the same strict rules, pilot120m keeps 36.9% attested words [E23, E24], about the subword models’ *unconstrained* level, with about twice as many distinct real words as their masking-only runs (757 against 323–375): each token is one akṣara, so the mask removes syllables, not multi-syllable subwords, and the model can still finish the words it starts. The base model has seen no poetry: 19.7% of its words are single akṣaras (real verse: 5.6%), and it repeats lines. Its *own* mode (Table 5) reaches 45.6% attested words but only 35 of 555 poems under the strict rules (313 under the relaxed profile).
- **A rule the decoders read differently.** meter\_engine reads any pāda-final laghu as guru. Under the treatise’s narrower reading (only before a conjunct), the Gemma poems need no edits, but 376 of the 555 masked pilot120m poems have edits at pāda ends (1,147 in all) [E25], and none under the engine’s reading. The pilot’s decoder relies on the lenient rule; the narrower one is an open choice (§8).

## 6.3 Evaluating what metre does not capture

**A deterministic poetics suite.** Following the *Kāvyaśaṅkārāsaṅgrahamū* (Rāmarājabhūṣaṇuḍu, 1920, 1945) and the Sanskrit sources it draws on, we built five metrics; Table 3 gives the definition, formula and rubric of each. Each has a frozen corpus baseline, a written rubric and a spec that traces every choice to a source.

**Validation.** The texts’ own examples are placed as the texts place them: the treatise’s anuprāsa example sits at the 97th percentile of reference poems; *mādhurya* puts every soft example above every harsh one (24 of 24 pairs; 23 without the run-load term); *ojas* orders all 8 dense/light pairs built from the texts’ examples (5 are contrasts a text states), and *prasāda* the 5 clear/obscure pairs; and the chandas distance finds the faults of the commentary’s two broken verses in the pādas the commentator names. Under the engine’s reading of a pāda-final laghu, it is zero exactly when meter\_engine accepts the poem (12,672 of 12,672 labelled poems) [E26].

| model                  | mode        | in<br>metre | attested     | distinct   | single<br>akṣara | strong<br>anuprāsa | obscure | MAUVE |
|------------------------|-------------|-------------|--------------|------------|------------------|--------------------|---------|-------|
| Gemma-4 E4B            | free        | 0           | 33.7%        | 490        | 0.4%             | 0.4%               | 2.2%    | 0.006 |
|                        | constrained | 555         | 10.7%        | 323        | 9.2%             | 33.7%              | 91.4%   | 0.057 |
| Gemma-4 26B-A4B        | free        | 0           | 37.2%        | 545        | 2.0%             | 9.9%               | 0.9%    | 0.009 |
|                        | constrained | 555         | 16.9%        | 375        | 16.1%            | 48.1%              | 91.0%   | 0.014 |
| DiffusionGemma 26B-A4B | free        | 0           | 34.4%        | 485        | 4.4%             | 19.1%              | 30.3%   | 0.006 |
|                        | constrained | 555         | 11.5%        | 333        | 28.7%            | 71.9%              | 98.9%   | 0.011 |
| pilot120m (ours)       | free        | 0           | 17.9%        | 183        | 5.9%             | 26.7%              | 45.9%   | –     |
|                        | constrained | 555         | <b>36.9%</b> | <b>757</b> | 19.7%            | 33.7%              | 62.9%   | –     |
| real verse             | –           | –           | –            | –          | 5.6%             | 2.7%               | 10.0%   | 0.965 |

Table 2: The constrained-decoding grid (555 poems per row; masking only for the Gemma models; the other strategies in Table 5). *Attested*: words of two or more akṣaras found in held-out verse (topic words excluded); *distinct*: distinct attested words; *strong anuprāsa*, *obscure*: extreme bands of the poetics metrics (Table 3); *MAUVE*: against real verse of the same metre mix ( $n = 120$  a side, 5 seeds; not run for pilot120m) [E22–E25] [E29].

| metric                            | definition                                                                                                                                                                                           | formula                                                                                                                                                                                                                | poem rubric                                                                                                               |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <i>anuprāsa</i><br>(alliteration) | the share of consonant pairs in a line that are the same sound, Simpson’s index of concentration (Simpson, 1949), as a z-score under a chance model with exact moments                               | $A$ same-sound pairs among $M$ consonant pairs in different akṣaras of a line; $E = qM$ , $q = \sum_s p_s^2$ over corpus frequencies; $z = (A - E)/\sqrt{V}$ , $V$ the exact chance variance                           | line levels at $z = 1.645, 2.326, 3.090$ (none, weak, marked, strong); poem score = mean line level                       |
| <i>mādhurya</i><br>(euphony)      | six classes of akṣara, from Mammaṭa’s soft and harsh lists (Mammaṭa, 2020) and the sonority scale (Parker, 2011), with a run-load term for harsh akṣaras that pile up                                | $w$ : madhura +1, sonorant +1/2, voiced 0, voiceless and conjunct −1/2, paraṣa −1; $M = I - P$ , $I = (1/N) \sum_t w_t$ , $P = (1/N) \sum_{t \text{ harsh}} R_{t-1}$ , $R$ the run load                                | harsh $< -0.283 \leq$ leaning harsh<br>$< -0.123 \leq$ balanced<br>$< 0.049 \leq$ leaning soft<br>$< 0.156 \leq$ soft     |
| <i>ojas</i> (density)             | the mean of the shares of akṣaras inside a word and opening with a conjunct, the features of Indic readability formulas (Sinha et al., 2012); no constant depends on a corpus                        | $O = (W + J)/2$ ; $W = 100(1 - 1/L)$ , the percentage of akṣaras that continue a word ( $L$ akṣaras per word); $J$ , the percentage that open with a conjunct                                                          | light $< 39.3$ (the treatise’s example of words apart, 4.61)<br>$\leq \dots < 47.5$ (its ojas example, 4.70) $\leq$ dense |
| <i>prasāda</i><br>(clarity)       | surprisal under an interpolated akṣara trigram model, the surprisal account of reading effort (Hale, 2001; Smith and Levy, 2013), trained on crawled poems disjoint from the evaluation files        | $H = -(1/N) \sum_t \log_2 P(a_t   a_{t-2}, a_{t-1})$ bits per akṣara; the mass kept for unseen akṣaras grows with the distinct akṣaras seen after a context (Witten and Bell, 1991)                                    | clear $< 5.12 \leq$ leaning clear<br>$< 5.54 \leq$ ordinary $< 6.33 \leq$ leaning obscure $< 7.06 \leq$ obscure           |
| chandas<br>distance               | the exact edit distance from a poem to every line its metre accepts, by dynamic programming over the automaton (Wagner, 1974; Mohri, 2003), plus yati and prāsa edits from the engines (Algorithm 4) | $D = \text{weight} + \text{yati} + \text{prāsa edits}$ ; $r = D / \max(n, N)$ , $n$ and $N$ the akṣaras of the poem and of the nearest valid one; $S = 1 - r/r_{\text{chance}}(m)$ , scaled by the metre’s chance rate | exact ( $D = 0$ ), slip, partial, weak, unmetred ( $r$ at or above the metre’s chance cut)                                |

Table 3: The poetics suite: the definition, formula and poem rubric of each metric. Each metric’s spec, baseline and validation are in poetry\_metrics/ [E26].

**The admission test.** The proposal admits a metric only if it declines monotonically when verse is degraded (word shuffle, metre breaking, filler, synonym swap), and requires metrically valid nonsense to be rejected by every meaning metric. The chandas distance declines under all four and is admitted [E27]. Prasāda declines under three and rises under the synonym swap, which substitutes plainer glosses, as a clarity metric should; it rejects the nonsense (200 of the 4,995 constrained poems rate “ordinary” or clearer [E25]) and is admitted. The sound metrics describe texture: anuprāsa declines under shuffle and swap, mādhurya and ojas ignore word order, and both sound metrics rise under filler (a line of ॐ is “strongly alliterative” and “soft”); they are read only behind the prasāda gate.

**MAUVE.** MAUVE (Pillutla et al., 2021), computed with the 300M EmbeddingGemma encoder (Schechter Vera et al., 2025), passed our gate on canonical Pothana (same-poet halves 0.957, structure-matched gibberish 0.005; Appendix Table 6) [E28]. Against real verse of the same metre mix (120 poems a side, 5 seeds), every generated set is at the floor (0.006–0.057, against 0.965 for real against real and 0.128 for akṣara-shuffled verse) [E29]; mean-pooled Gemma-3-1B states (Gemma Team, 2025) agree (0.005–0.014). MAUVE reads register and vocabulary, not order or metre: word-shuffled verse still scores 0.91, and Pothana’s own prose 0.015. It cannot tell fluent modern Telugu (the free baselines) from metrically valid nonsense, which *prasāda* separates (5.4–6.7 against 10.6–12.6 bits per akṣara), and the differences between cells

(0.01–0.05) are of the order of the seed-to-seed spread. We keep it only as a floor check. A Gemma-3-1B perplexity judge fails on verse too: it separates real from word-shuffled prose (16.9 against 64.9), but rates real verse at 113, shuffled verse at 150, and repetitive output better than both [E30].

## 7 Discussion and Limitations

**What is shown.** Metrical validity in Telugu can be decided exactly and enforced inside the decoding loop of very different generators; pretrained models fail through their units and their treatment of weight, not through ignorance of the rules; and with akṣara-level units the hard constraint stops destroying vocabulary.

**Śabda and artha.** A padyam is judged by the balance (*samanvaya*) of śabda and artha, sound and meaning. Up to this mid-point we have concentrated on the śabda side: the metre (*chandas*) and the sound of the line. The artha side is next, and this is where a word-resolution metric comes in: how much of a poem resolves into real words. Until such a metric and a coherence metric are in place, the śabda metrics mean little, because repetitive gibberish of consonants games them (a line of ష is “strongly alliterative” and “soft”, §6.3). We will build both on a macro dictionary and add lexical gates on the same basis; besides fine-tuning, they are what is meant to bring coherence (§8).

**What is not shown yet.** *No model writes good padyams:* our base model has seen no poetry, and its constrained output is fragmented and repetitive. *The revision hypothesis is untested:* the ability to revise a choice motivates diffusion, but our sampler fills positions in any order and never revises one. *Meaning has not been measured by people:* every automatic metric here measures form, sound or familiarity. *The tools have limits:* the yati engine is the weakest on canonical text, because sandhi is unwritten; five-pāda forms are not accepted; and engine-assigned labels (Vemana, Kuchimanchi) cannot validate the engine.

## 8 Remaining Work and Timeline

The next phase, to the end of the project on 1 November 2026, adapts the diffusion architecture to poetry, fine-tunes the models exhaustively, and measures meaning with people. Every stage has a gate; if one is missed, the study proceeds with

the last model that passed. G1: poem NELBO below the base model’s, general NELBO up by under 2%; G2: the meaning lowers the poem’s NELBO; G3 (200 validation meanings): metre  $\geq$  99%, prāsa and yati within 5 points of real poems,  $\leq$  3% non-words; G4: the meaning score rises within the copy and diversity guards.

1. **2–7 Oct, mid submission:** the human-study interface, rubric and rater recruitment; the test split frozen.
2. **5–14 Oct, architectural adaptation** (G1, G2; G3): continued pretraining on poem + meaning documents ( $\approx$ 4 h); multi-task fine-tuning (meaning  $\rightarrow$  poem, with explainer, gloss and infill tasks) on a canvas with the meaning, a metre header and a register tag ( $\approx$ 13–15 h with a learning-rate sweep); a revision test against a remasking sampler (Wang et al., 2025) with validator-triggered remasking; and the rule choices (yati strictness, the pāda-final laghu) on validation data.
3. **5–18 Oct, word resolution and coherence (artha):** both metrics built on a macro dictionary and admitted by the suite’s test before the automatic evaluation, and lexical gates on the same dictionary.
4. **12–21 Oct, exhaustive fine-tuning** (G4): ablations; rejection-sampling self-training with the engines as verifiers ( $\approx$ 5–6 h a round); baselines: retrieval (valid by construction), a local few-shot LLM with engine filtering, a matched-size autoregressive model (to isolate diffusion), and the constrained Gemma grids; the 523M scale-up if G3 passes; the model table on the test split.
5. **19–24 Oct, automatic evaluation** of every system on the held-out Bhāgavatamu test split, each measure against the real-poem ceiling and the shuffled floor: the final automatic results.
6. **20–29 Oct, human evaluation** of 500 poems stratified by metre and system on an eight-item rubric (Appendix B), with at least two raters per poem and three on a 100-poem overlap: the ratings, their agreement, and the metric–item correlations that complete the admission rule.
7. **26 Oct–1 Nov, final report:** the final paper (1 November) and the release of the controller, corpora, suite and ratings.



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## A Algorithms

**Terms used in the algorithms.** An *akṣara* is a syllable; it is *guru* (heavy,  $\mathbf{u}$ ) or *laghu* (light,  $\mathbf{i}$ ). A *gaṇa* is a fixed group of weights, such as *bha* =  $\mathbf{u}\mathbf{i}\mathbf{i}$ . A poem has *pādas* (lines), and a *slot* is a kind of line that a metre allows: *kandamu*’s odd and even lines are two slots. At a *yati* seat, the *akṣara* must agree in sound (*maitri*) with the first *akṣara* of the line, the *vaḷi*. By *prāsa*, the second *akṣara* of every line repeats the consonant of the second *akṣara* of line 1. An *automaton* is a graph of *states* joined by edges marked  $\mathbf{u}$  or  $\mathbf{i}$ ; a line is legal when its weights lead from the start state to a *final* state. A *token* is the unit a language model emits: one *akṣara* for our model, a subword of one or more *akṣaras* for the Gemma models.

### Algorithm 1 Build the labelled metrical DAWG

**Input:** the metre catalogue and the *gaṇa* classes (YAML files)  
**Output:** the DAWG  $D$  (1,431 states, 2,514 edges; acyclic, as every line has a bounded length), and the label set  $C(q)$  of each state  $q$   
**Symbols:**  $m, s$ : a metre and one of its line slots (a *vr̥tta* has one slot; *kandamu* has two, its odd and even lines).  $\mathbf{u}, \mathbf{i}$ : *guru*, *laghu*.  $N_{m,s}$ : the automaton of the slot, whose *label* is  $(m, s)$ .  $C(q)$ : the labels whose lines can still be completed from state  $q$ .

- 1: **for** each metre  $m$  and each slot  $s$  of  $m$  **do**
- 2:   write the slot’s grammar: one position per *gaṇa*, each allowing the *gaṇas* of its class
- 3:   split every rule into rules that read one weight ( $\mathbf{u}$  or  $\mathbf{i}$ ) each
- 4:   turn the grammar into an automaton  $N_{m,s}$ , and label its final state  $(m, s)$
- 5: **end for**
- 6: join all the  $N_{m,s}$  into one automaton and make it deterministic (subset construction); a merged state keeps every label of the states it merges
- 7: remove the states that cannot reach a final state
- 8: put the states with the same labels into one group
- 9: **repeat**
- 10:   split a group whose states go to different groups on  $\mathbf{u}$  or on  $\mathbf{i}$  (Moore)
- 11: **until** no group splits
- 12: merge each group into one state; the result is  $D$
- 13: for each state  $q$ , from the ends of the lines back to the start:  $C(q) \leftarrow$  the labels of  $q$  together with  $C$  of each state that  $q$  leads to
- 14: **return**  $D$  and  $C$

### Algorithm 2 The enforcer

**Input:** the metre  $M$ ; the state  $\sigma$  of the text so far; the next *akṣara*  $a$  (or a space or a line break)  
**Output:** the new state, or DEAD if no poem of  $M$  can begin with the text  
**Symbols:** The state  $\sigma$  has four parts.  $Q$ : the states of  $M$ ’s automaton (built as in Algorithm 1, for  $M$  alone) that the text may be in.  $w$ : the last *akṣaras*, whose weight is still open (a light *akṣara* before a conjunct is heavy). The *prāsa register*: the consonant of line 1’s second *akṣara*, which the second *akṣara* of every line must repeat. The *yati register*: the line’s first *akṣara* (the *vaḷi*), with which the *akṣara* at a *yati* seat must agree in sound (*maitri*).

- 1: settle the open weights in  $w$  by the first consonant of  $a$
- 2: move each state in  $Q$  along the settled weights; drop those with no move
- 3: **if**  $a$  is the second *akṣara* of a line and breaks the *prāsa* **then return** DEAD
- 4: drop from  $Q$  the states whose *gaṇa* division puts  $a$  on a *yati* seat where  $a$  is not in *maitri* with the *vaḷi*
- 5: **if** no state in  $Q$  can still reach the end of the poem **then return** DEAD
- 6: **return**  $\sigma$  with  $Q$ ,  $w$  and the registers updated

A token is *allowed* when the state stays alive after all of its *akṣaras* (cached per state). Liveness is exact, so decoding never reaches a dead end, and every finished poem is valid. The *gaṇa* grid acts as a mask from the first token; *yati* and *prāsa* act once their anchor (the *vaḷi*, line 1’s *prāsa akṣara*) is written. Meaning has no closed-form check: it is left to the generator and the evaluation.

### Algorithm 3 Decoding with three strategies

**Input:** a language model; a metre  $M$ ; a strategy (masking only, backtracking or hybrid); a temperature  $\tau_0$   
**Output:** a poem  $y$  that is valid for  $M$   
**Symbols:**  $y$ : the poem so far;  $\sigma$ : its enforcer state (Algorithm 2);  $A$ : the allowed tokens;  $p$ : the model’s probabilities for the next token;  $\tau$ : the temperature (probabilities are raised to the power  $1/\tau$ );  $c$ : the number of backtracks in a row; a *checkpoint*: a saved copy of  $(y, \sigma, \tau)$ . The run is *stuck* when fewer than 3 tokens are allowed, when they hold less than 0.001 of  $p$ , or after 4 steps without a new *akṣara*.

- 1:  $y \leftarrow$  the empty poem;  $\sigma \leftarrow$  the start state;  $\tau \leftarrow \tau_0$ ;  $c \leftarrow 0$
- 2: **while**  $y$  is not a complete poem **do**
- 3:   **if** the line is complete and cannot grow **then** add a line break, and go to the next step
- 4:   backtracking: every third step, save a checkpoint (keep the last 15)
- 5:    $A \leftarrow$  the allowed tokens;  $p \leftarrow$  the model’s probabilities
- 6:   **if** backtracking and the run is stuck **then**
- 7:      $c \leftarrow c + 1$ ; go back  $c$  checkpoints; raise  $\tau$  by 0.15 $c$ , to at most 1.2; go to the next step
- 8:   **end if**
- 9:   mask: set  $p$  to 0 outside  $A$ , apply  $\tau$ , and renormalise
- 10:   **if** hybrid **then**
- 11:     go down the allowed tokens by probability (at most 100), collecting those that complete the line; stop at 3, or at the 10th token if none was found; sample among the collected tokens, or else among those 10
- 12:   **else**
- 13:     sample from the most probable tokens that together hold 90% of  $p$  (top- $p$ )
- 14:   **end if**
- 15:   add the token to  $y$ ; update  $\sigma$ ;  $c \leftarrow 0$
- 16: **end while**
- 17: **return**  $y$

After a backtrack,  $\tau$  returns to  $\tau_0$  over 6 steps; the autoregressive runs use  $\tau_0 = 0.7$ . *Block diffusion* commits the model’s own proposals left to right while each is allowed and confident; otherwise the strategy chooses the token at the first open position, and a backtrack returns committed tokens to noise.

### Algorithm 4 Chandas distance

**Input:** the printed lines of a poem, each *akṣara* with the weights it may take; a metre  $m$   
**Output:** the distance  $D$ , and the skill  $S$  (1 for a poem in metre, 0 for one no nearer to  $m$  than prose)  
**Symbols:**  $x$ : a printed line, and  $x_i$  its  $i$ -th *akṣara*;  $q$ : a state of the slot’s own automaton (built as in Algorithm 1);  $\text{cost}[i][q]$ : the fewest edits that read  $x_1 \dots x_i$  and end in state  $q$ ;  $n, N$ : the number of *akṣaras* in the poem and in the nearest valid poem;  $r$ : the edit rate;  $r_{\text{chance}}(m)$ : the mean rate of prose cut to the line lengths of  $m$ .

- 1: **for** each printed line  $x$  and each slot  $s$  of  $m$  **do**
- 2:   fill  $\text{cost}[i][q]$  for  $i = 0, 1, \dots$ : reading  $x_i$  along an edge costs 0 if the edge’s weight is one that  $x_i$  may take, else 1 (substitution); skipping  $x_i$  costs 1 (deletion); following an edge without reading costs 1 (insertion)
- 3:   the line’s distance is the lowest cost at a final state after the last *akṣara* (a *pāda*-final *laghu* may also read as  $\mathbf{u}$ ); its path gives the nearest legal line
- 4: **end for**
- 5: weight edits: match the lines to the metre’s *pādas* with the fewest edits; a line left over costs its length, a *pāda* left empty its shortest line
- 6: *yati* edits: one for each *yati* seat whose *akṣara* is not in *maitri* with the *vaḷi*
- 7: *prāsa* edits: the fewest *pādas* to change so that all the *pādas* rhyme
- 8:  $D \leftarrow$  the sum of the three, counting once an *akṣara* that a weight edit already rewrites
- 9:  $r \leftarrow D / \max(n, N)$ ;  $S \leftarrow 1 - r / r_{\text{chance}}(m)$
- 10: **return**  $D$  and  $S$

## B Additional Tables

**Pilot 1** (one-shot poems from Gemini 3.7 Flash, scored by our engines; strict profile, compound readings excluded): 91 of 255 (35.7%) are valid. Valid / poems by metre: *utpalamāla* 28 / 53, *campakamāla* 17 / 52, *śārdūla* 4 / 46, *mattēbha* 4 / 40, *āṭaveladi* 10 / 13, *tēṭagīti* 9 / 13, *sīsamu* 0 / 13, *kandamu* 8 / 13, *utsāhamu* 11 / 12 [E1].

| check                                             | result                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| automaton and tests [E7, E10]                     | 37 metres, 40 slots; 1,431 states, 2,514 edges; 370,315 legal lines; acyclic, minimal; 541 tests pass, including grammar $\leftrightarrow$ automaton equivalence and a brute-force identification oracle                                                                                                                                   |
| identification, Bhāgavatamu (7,365 labelled) [E8] | 7,355 as labelled (99.86%); 0 as a wrong metre; 10 none                                                                                                                                                                                                                                                                                    |
| prāsa, Bhāgavatamu (4,545 poems) [E3]             | 96.7% strict, 99.8% relaxed                                                                                                                                                                                                                                                                                                                |
| yati, Bhāgavatamu (7,355), by sandhi policy [E11] | strict, none 63.6%; strict, hypothesis 90.7%; relaxed, hypothesis 91.4%; relaxed, vowel-only 99.8%                                                                                                                                                                                                                                         |
| Chandassu annotation (4,651 poems) [E9]           | segmentation identical for 74.3% (1,036 of the 1,196 differences begin at a word-final dead consonant, which the engine attaches to the preceding akṣara and the annotation to the next); weights of the 227,357 akṣaras of identically segmented poems: 99.10% agree with the engine’s canonical reading, 99.76% within its open readings |

Table 4: The controller and its validation.

| model                  | mode         | in<br>metre      | attested | distinct | single<br>akṣara | strong<br>anuprāsa | obscure | MAUVE |
|------------------------|--------------|------------------|----------|----------|------------------|--------------------|---------|-------|
| Gemma-4 E4B            | backtracking | 555              | 12.2%    | 394      | 9.3%             | 23.1%              | 83.8%   | 0.037 |
|                        | hybrid       | 555              | 11.7%    | 355      | 7.9%             | 30.3%              | 89.4%   | 0.035 |
| Gemma-4 26B-A4B        | backtracking | 555              | 20.8%    | 525      | 12.0%            | 49.0%              | 83.1%   | 0.015 |
|                        | hybrid       | 555              | 16.7%    | 436      | 14.1%            | 47.7%              | 87.2%   | 0.027 |
| DiffusionGemma 26B-A4B | backtracking | 555              | 11.6%    | 432      | 25.0%            | 46.7%              | 98.9%   | 0.032 |
|                        | hybrid       | 555              | 12.9%    | 330      | 26.5%            | 78.9%              | 93.7%   | 0.011 |
| pilot120m (ours)       | own          | 35 (313 relaxed) | 45.6%    | 1,451    | 17.8%            | 20.5%              | 51.7%   | –     |

Table 5: The other decoding strategies on the grid (555 poems each); free and masking-only runs are in Table 2, whose columns these are. *Own*: the pilot’s own mode (soft yati penalty, 16 particles, soft lexicon), counted under the strict rules [E22–E25, E29].

| condition                                                  | MAUVE             |
|------------------------------------------------------------|-------------------|
| held-out Pothana vs Pothana (1,601)                        | 0.957 $\pm$ 0.003 |
| line order shuffled (1,601)                                | 0.906 $\pm$ 0.007 |
| word order shuffled (1,601)                                | 0.716 $\pm$ 0.034 |
| vṛtta vs jāti, both Pothana (721)                          | 0.214 $\pm$ 0.029 |
| akṣaras shuffled (1,601)                                   | 0.028 $\pm$ 0.002 |
| Pothana vs Vemana, āṭaveladi only (313)                    | 0.051 $\pm$ 0.007 |
| structure-matched gibberish (1,601)                        | 0.005 $\pm$ 0.000 |
| real vs real (120)                                         | 0.965 $\pm$ 0.006 |
| real vs its words shuffled (120)                           | 0.912 $\pm$ 0.065 |
| real vs its akṣaras shuffled (120)                         | 0.128 $\pm$ 0.042 |
| real verse vs Pothana’s prose (120)                        | 0.015 $\pm$ 0.004 |
| generated poems, 12 model $\times$ decoding cells (120)    | 0.006–0.057       |
| the same 12 cells, Gemma-3-1B mean-pooled featurizer (120) | 0.005–0.014       |

Table 6: MAUVE with embeddinggemma-300m. Top: the gate on Pothana (5 seeds;  $n$  per side). Bottom: controls of the generated-poem comparison (real verse with the matched metre mix,  $n = 120$ , 5 seeds); for the Gemma-3-1B featurizer, real against real scores 0.940. The generated cells are in Table 5 [E28, E29].

### The human-evaluation rubric (planned; §8).

(1) How easy is it to read? (1–5) (2) How easy is it to understand immediately after reading? (1–5) (3) How many words of the poem are understood? (1–5) (4) Is the chandassu correct? (0/1) (5) Is the prāsa correct? (0/1) (6) Is the yati correct? (0/1) (7) Does the poem express the given meaning? (1–5) (8) How good is it as a poem? (1–5). Items 1–6 come from the proposal; items 7–8 are added for the study. Three 500-poem pools of engine-verified real and generated poems are ready for it.

The tables are generated from the engine’s rule files by `paper/scripts/make_rule_tables.py`.

Table 7: The 37 metres of the catalogue (meter\_rules.yaml). *Vṛtta gaṇas*: bha uii, ja iui, sa iuu, na iii, ya iuu, ra uui, ta uui, ma uui, ga u, va iu, gaga uu. *Sūrya gaṇas*: na, ha (ui); indra gaṇas: bha, ra, ta, na, naga, sala. *Yati*: the seat that must agree with the first akṣara (of the pāda, or of gaṇa 1); PY: prāsa-yati, a prāsa match at the seat, is accepted in place of yati. *Lines*: the legal weight strings of a pāda (odd / even pādas; + 1 for the all-laghu sāsamu); a vṛtta has one. Bhāgavatamu poems by label (level-0 inventory): kandaṃ 2,608, tēṭṭai 1,051, sāsamu 1,048, āṭavelai 716, mattēbhavikṛḍitaṃ 586, campakamāla 490, utpalamāla 471, śārdūlavikṛḍitaṃ 289, mattakōkilaṃ 41, taralaṃ 23, māliṇi 12, and 30 in 15 other metres.

Table 8: The *prāsa* rules (*prasa\_rules.yaml*), distilled from our notes on the classical treatises, among them the *Appakavīyam* (Appakavi, 1966) (64 rules). Status: C canonical, Cs canonical subtype, M mandatory (all profiles), R accepted relaxation (relaxed and historical profiles), O orthographic relaxation, D deprecated (historical only), F forbidden (a named failure), I informational.

Table 9: The yati rules (`yati_rules.yaml`), distilled from chapter 6 of Padya Vidya (Peri, 2018) (92 rules). Status: C canonical, Cs canonical subtype, M mandatory (all profiles), R accepted relaxation (relaxed and historical profiles), O orthographic relaxation, D deprecated (historical only), F forbidden (a named failure), I informational.

## D Results of the Experiments

The text gives the results of these experiments in words; this appendix gives their numbers (Tables 10–13) and their curves (Figure 3). A tag in the text links to the results of its experiment: E1 to Appendix B, E2–E5 to Table 1, E7–E11 to Table 4, E6, E12–E21 and E30 to Table 10, E22–E25 to Tables 2 and 5, E26 to Table 13, E27 to Table 12, and E28–E29 to Table 6.

| experiment | measure                                                              | result                                                                                   | baseline or control                                                                                                                                                                                                        |                                                                                                           |
|------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| E6         | duplicates in the four files                                         | repeated verse among the 12,704 poems; pādas shared between poems                        | no exact duplicate; 2 pairs equal after normalisation and 6 near pairs, none across files; a pāda shared with another poem: Vemana 77.1% (its refrain, in 893 poems), Chandassu 25.3%, Kuchimanchi 11.8%, Bhāgavatamu 1.1% | near: character 3-gram Jaccard $\geq 0.9$                                                                 |
| E12        | pretraining pilot120m, a masked diffusion model (Sahoo et al., 2024) | test NELBO in nats per token, on 512 canvases per source                                 | Sangraha 1.846, IndicCorp v2 1.729, Wikipedia 1.053 (Khan et al., 2024; Doddapaneni et al., 2023); 1.783 weighted as in training                                                                                           | validation split: 1.818, weighted the same way                                                            |
| E13        | akṣara coverage of two tokenizers                                    | the 3,612 akṣara types (897,463 occurrences) of our verse that have a token of their own | Gemma 4 (Gemma Team, 2026): 475 single-akṣara tokens, 10.6% of the types, 81.9% of the occurrences; ours: 99.8% and over 99.99%                                                                                            | –                                                                                                         |
| E14        | teacher-forced surprisal                                             | mean surprisal of the true token (nats); share of tokens the model ranks first           | verse 6.40, its meaning 5.35; the meaning is easier for 186 of 200 poems; ranked first: 16.3% of verse tokens, 32.3% of meaning tokens                                                                                     | without <bos>: verse 13.45, above $\ln  V  = 12.48$ ; random weights: 12.80                               |
| E15        | free generation, greedy                                              | poems in a metre; akṣaras with the template weight                                       | 0 of 200 in any metre (172 have four lines); within the template’s length, 50.8% of 5,550 steps have the wrong weight; template weight 68.4% teacher-forced, 51.0% free                                                    | best layer: the embedding layer, 84.7%; no layer beats the lookup                                         |
| E16        | linear probe for weight                                              | accuracy (5-fold) on the 3,938 tokens whose weight the next akṣara sets                  | majority class and lookup of the akṣara 88.7%; lookup with the next onset 99.1%                                                                                                                                            |                                                                                                           |
| E17        | subword boundaries in verse                                          | Gemma token boundaries that fall inside an akṣara                                        | 15.5% (Vemana) to 21.8% (Bhāgavatamu); 1.23–1.36 tokens per akṣara                                                                                                                                                         | the meanings: 14.4–19.6%; 0.93–0.98 tokens per akṣara                                                     |
| E18        | register probe                                                       | accuracy (5-fold), verse against its meaning, 400 texts                                  | 0.588 at layer 0; 0.978 at layer 1; at least 0.98 from layer 2; 1.000 first at layer 24                                                                                                                                    | chance 0.5; text length alone 0.565                                                                       |
| E19        | metre geometry against gaṇa distance                                 | Spearman r between activation distance and gaṇa distance over the 28 metre pairs         | $p < 0.05$ at 34 of 36 layers; peak 0.773 at layer 16 (centred directions: 33 layers, 0.745 at layer 10)                                                                                                                   | Mantel test, 999 permutations                                                                             |
| E20        | metre directions                                                     | coherence of difference-in-means directions                                              | verse — meaning: 0.475 (layer 35); the 28 metre pairs: 0.003–0.158 (median 0.064), none at the usable 0.322                                                                                                                | random splits: 95th percentile 0.068, 99th 0.128 (13 pairs above the 95th)                                |
| E21        | tight contrast set                                                   | coherence of the metre-only difference over 357 triples                                  | at most 0.011 (layer 0)                                                                                                                                                                                                    | sign-flip null, 95th percentile 0.034; breaking and keeping edits point the same way (median cosine 0.72) |
| E30        | perplexity judge, gemma-3-1b-pt (Gemma Team, 2025)                   | perplexity of real and generated text                                                    | prose 16.9, its words shuffled 64.9; verse 113.3, its words shuffled 149.7; generated poems 3.5–159.8, lowest for the pilot’s free output, where 212 of 555 poems repeat a line                                            | 30 prose texts; 555 Chandassu poems                                                                       |

Table 10: Experiments whose results the text gives in words. E14–E21 use gemma-4-E2B-it (Gemma Team, 2026) on the 200 Bhāgavatamu poems of §6.1 and their meanings.

| hypothesis                                                                                         | statistic (95% CI)        | $p$                 | verdict       |
|----------------------------------------------------------------------------------------------------|---------------------------|---------------------|---------------|
| the surprisal gap (true token — the model’s choice) is larger for verse than for its meaning [E14] | +0.50 nats (0.39 to 0.61) | $5 \times 10^{-15}$ | holds         |
| the gap concentrates at pāda starts [E14]                                                          | −0.31 (−0.57 to −0.06)    | 0.99                | opposite sign |
| prāsa seats get easier once pāda 1 fixes them [E14]                                                | +0.39 (−1.10 to 1.90)     | 0.69                | no            |
| word-initial yati seats are easier (61 poems) [E14]                                                | +1.05 (0.24 to 1.92)      | 0.97                | opposite sign |
| the template weight is kept less when free [E15]                                                   | 68.4% against 51.0%       | $1 \times 10^{-9}$  | holds         |

Logistic regression of a wrong weight within the template (5,550 steps, standardised, cluster-robust) [E15]: rare token +0.57 ( $p = 2 \times 10^{-7}$ ); entropy +0.04 (0.19); akṣaras since a word junction −0.03 (0.37); position in the pāda +0.08 (0.02); absolute position −0.02 (0.37).

Table 11: Tests of the reading and generation diagnostics.  $p$ : one-sided Wilcoxon over poems, except the fifth row (two-sided, over 93 metre slots). A rare token occurs fewer than 10 times in the tokenised Bhāgavatamu.

| metric           | words shuffled | metre broken | filler | synonyms swapped | nonsense above real  |
|------------------|----------------|--------------|--------|------------------|----------------------|
| chandas distance | ↓              | ↓            | ↓      | ↓                | in metre by design   |
| prasāda          | ↓              | ↓            | ↓      | ↑                | 0.005–0.08: rejected |
| anuprāsa         | ↓              | ↑            | ↑      | ↓                | 0.68–0.95            |
| mādhurya         | ○              | ↓            | ↑      | ↓                | 0.25–0.53            |
| ojas             | ○              | ○            | ↓      | ↑                | 0.62–0.74            |

Table 12: The admission test [E27]: 1,600 real poems (400 per file), each degradation at four strengths. ↓: the mean falls at every strength and a sign test at full strength has  $z \leq -3.09$ ; ↑:  $z \geq 3.09$ , the metric rewards the degradation; ○: neither. *Nonsense above real*: the share of pairs in which one of the 4,995 constrained Gemma poems scores above a real poem, over the 9 model–strategy cells.



| metric           | the texts' examples                                                                                                                        | result                                                                                                                 | baseline                     |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------|
| anuprāsa         | the treatise's example of <i>vr̥tṭyanuprāsa</i> , 4.77 (Rāmarājabhūṣaṇuḍu, 1920)                                                           | $var̥ṇa\ z = 3.18$ , the 97th percentile of reference poems; poem score at the 99th percentile                         | –                            |
| mādhurya         | 4 soft and 6 harsh examples from the treatise, its commentary (Rāmarājabhūṣaṇuḍu, 1945), Mammaṭa (Mammaṭa, 2020) and Daṇḍin (Daṇḍin, 2020) | every soft example above every harsh one: 24 of 24 pairs; 23 without the run-load term                                 | sonority alone: 17 of 24     |
| ojas             | 8 dense/light pairs built from the treatise, Mammaṭa, Daṇḍin, Vāmana (Vāmana, 2020) and the commentary (5 are contrasts a text states)     | 8 of 8 in order                                                                                                        | with aspirates added: 6 of 7 |
| prasāda          | 5 clear/obscure pairs: the treatise's clear diction against four of its faults (4.5, 4.8, 4.15, 4.17), and a pair from Daṇḍin              | 5 of 5 in order                                                                                                        | –                            |
| chandas distance | the commentary's two broken verses; the 12,672 labelled poems of the four files                                                            | the faults found in the pādas the commentator names; zero exactly when meter_engine accepts the poem, 12,672 of 12,672 | –                            |

Table 13: The poetics suite on its texts' own examples [E26]. Verse numbers are those of the treatise.

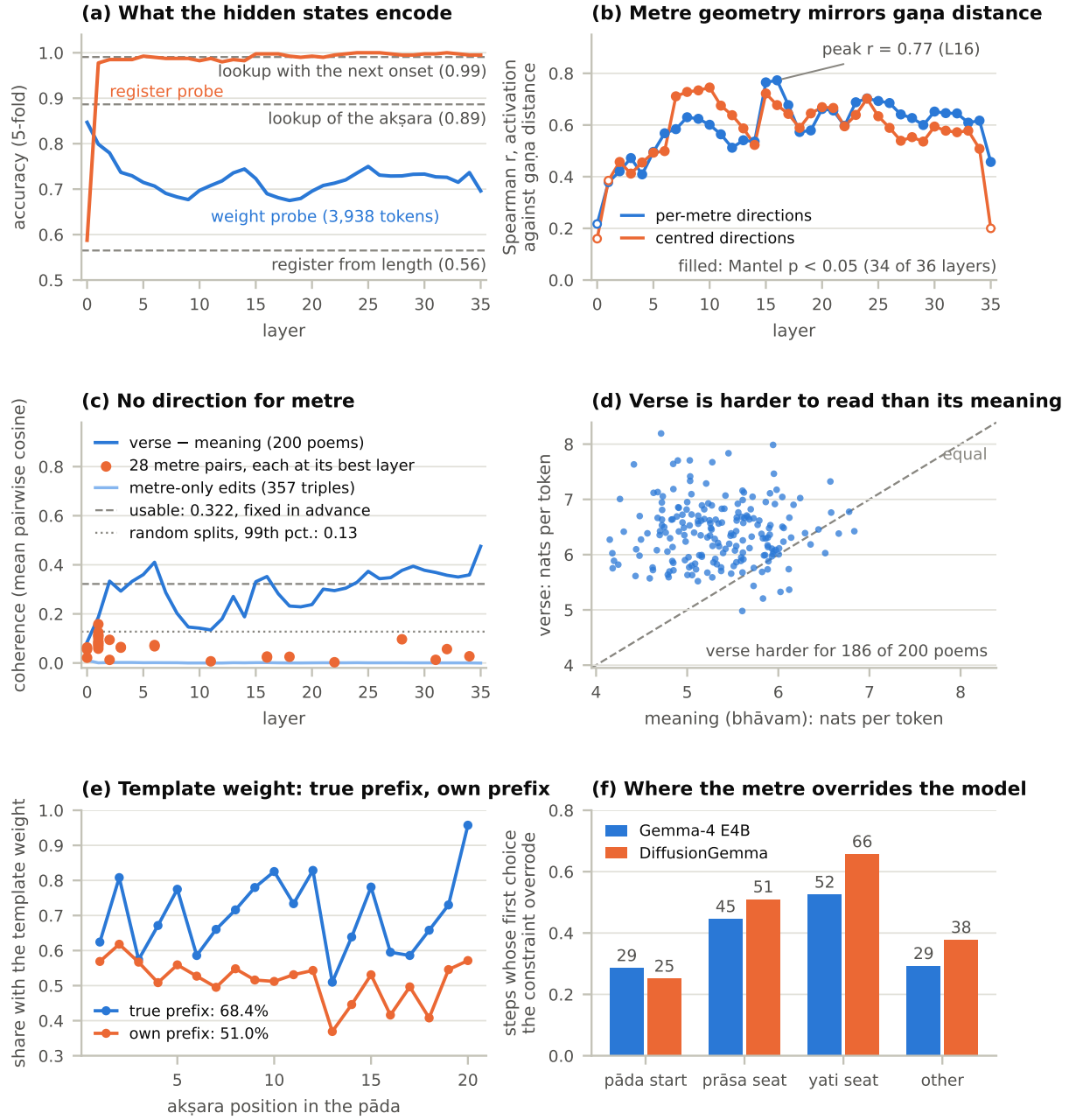


Figure 3: The experiments of Table 10 as curves. (a) Accuracy of the weight probe on the 3,938 tokens whose weight the next akṣara sets [E16], and of the register probe [E18], by layer, against their baselines. (b) Spearman  $r$  between activation distance and gaṇa distance by layer; filled points have a one-sided Mantel  $p < 0.05$  [E19]. (c) Coherence of the verse — meaning direction by layer, the best layer of each of the 28 metre pairs [E20], and the metre-only difference of the tight contrast set [E21]. (d) Mean surprisal per poem of the verse against its meaning, with <bos> [E14]. (e) The share of akṣaras with the template weight by position in the pāda, with the true prefix and with the model's own, in the five fixed metres [E15]. Panels a–e use gemma-4-E2B-it (Gemma Team, 2026). (f) The share of decoding steps whose first choice the constraint overrode, by position, over the three strategies, for Gemma-4 E4B (Google DeepMind, 2026b) and DiffusionGemma (NVIDIA, 2026) [E22].